

3.2 Menus

There are five menu options found at the top of the main window: **File**, **Setup**, **Modify Datasets**, **Tools** and **Help**.

- **File.** This menu provides options for selecting saved project files (*.projx*) to open, starting a new project file, saving your work in a project file, and exiting the program.
- **Setup.** BenMAP-CE comes pre-loaded with datasets for the *United States*, *China*, and *Detroit* setups. The selected setup will be displayed in the menu bar, next to the File menu. You may click on the selected setup to see a menu of available setups. Each setup includes the information needed to run analyses for a particular geographic area. To learn more about modifying setups, see Chapter 4: Loading Data.
- **Modify Datasets.** BenMAP-CE stores the information needed to run analyses for a particular geographic area, such as a city, region, or nation in a single dataset called a setup. Many users will never need to modify the preloaded setups. However, the **Modify Datasets** menu provides tools to add, modify (load additional datasets), or delete these setups if needed. This is discussed in detail in Chapter 4: Loading Data.
- **Tools.** This menu provides access to data import and export functionality in addition to a number of other features. An overview is provided below in Section 3.2.1.
- **Help.** This menu provides access to a *Quick Start Guide* (available on EPA's website), information *About* BenMAP-CE, and a form to *Provide Feedback* about software errors or requested features. An overview is provided below.

3.2.1 Tools Menu

The **Tools** menu has several options: *Aggregate Air Quality Surface*, *Database Export*, *Database Import*, *Online Database Export*, *Online Database Import*, *Export Air Quality Surface*, *GBD Rollback*, *Monitor Data Conversion*, *Neighbor File Creator*, *PopSim Options*, and *Compute Grid Crosswalks*. A brief description is given below, but further information can be found in Chapter 9: Tools Menu.

- *Aggregate Air Quality Surface.* Create a new air quality surface for a specified grid definition (e.g., County) from an existing air quality surface created with a different (and generally finer) grid definition (e.g., 12km CMAQ).
- *Database Export.* Export all or part of BenMAP-CE's internal database to a database (*.bdbx*) file or multiple csv or shapefiles which can later be used on another computer or by another user. Manually loading data into BenMAP-CE can be time and labor intensive, so this tool can be quite useful in sharing data with other users

or computers. This tool can be used to share all or part of an existing setup, or to backup Setups prior to upgrading BenMAP.

- *Database Import.* Import data created using the *Database Export* tool into a specified setup.
- *Online Database Export* (Feature currently disabled). Allows users to post/share data for BenMAP-CE via a cloud-based system.
- *Online Database Import* (Feature currently disabled). Allows users to download data shared by others for BenMAP-CE via a cloud-based system.
- *Export Air Quality Surface.* Generate a text file (.csv) with all of the data in air quality surface, including summary statistics such as mean, median, minimum, and maximum.
- *GBD Rollback Tool.* The *GBD Rollback* tool allows you to select a country, region, or group of countries and see the impact of lowering PM_{2.5} emissions based on the data from the 2010 GBD study¹². The outputs include the baseline and policy case PM_{2.5} concentrations as well as the population-weighted air quality change.
- *Monitor Data Conversion.* Create a monitor file in BenMAP-CE format using an external data file in a different format.
- *Neighbor File Creator.* This tool creates a text file (.txt) identifying "neighbor" monitors and associated interpolation weights for each grid cell in an air quality grid. Identifying the major contributors to air quality in a given cell can be helpful for understanding and troubleshooting BenMAP results, and for understanding how different geographic patterns of air quality changes can affect health benefits.
- *PopSim.* A dynamic population simulation that incorporates the cumulative effects of air pollution on different age groups over time.
- *Options.* Select options for start-up and exit screens, validation logs and default setups.
- *Compute Grid Crosswalks.* Remove all crosswalks for selected setups and re-create them. This tool is for creating or repairing crosswalks which were broken due to database error or force quitting of the application.

¹² Exposure Assessment for Estimation of the Global Burden of Disease Attributable to Outdoor Air Pollution. Michael Brauer, Markus Amann, Rick T. Burnett, Aaron Cohen, Frank Dentener, Majid Ezzati, Sarah B. Henderson, Michal Krzyzanowski, Randall V. Martin, Rita Van Dingenen, Aaron van Donkelaar, and George D. Thurston. *Environmental Science & Technology* **2012** 46 (2), 652-660

3.2.2 Help Menu

The **Help** menu has a few options to choose from: *Quick Start Guide link*, *User Documentation*, *About*, and a form to *Provide Feedback*.

- *Quick Start Guide.* A link will open a webpage to the U.S. EPA site for BenMAP-CE training materials. On the webpage, users can download a series of self-paced exercises for seven different regions of the world, including two exercises in Spanish and one in French. Each self-paced exercise will take you through the basic operations of BenMAP-CE data imports and decision-making.
- *User Documentation.* Opens a link to the U.S. EPA site with the BenMAP-CE user manual and appendices.
- *About.* Opens a window that displays information about the program (e.g., software version, contact information, and a suggested citation). You can click on **Release Notes** to read about software modifications and any known issues.
- *Provide Feedback.* This feature allows you to submit any problems that you may encounter while running BenMAP-CE or requested features to the BenMAP-CE development team. There are fields to provide your contact information (optional) and information about the error or requested feature. Your feedback will be logged into an issue tracking system for U.S. EPA to evaluate.

Provide Feedback

BenMAP-CE is open-source software. If you encounter an error or would like to request a feature, please consider submitting a report by filling out this form. We apologize for any issues you may have encountered and value your feedback as it will help us continue to improve BenMAP-CE.

Operating System: Microsoft Windows NT 6.1.7601 Service Pack 1

BenMAP-CE Version: 1.3.5.12 (April 18, 2017)

Your name (optional):

Your email address (optional):

Country where you are located (optional):

What type of report would you like to submit?

☒ Software Error

☐ Requested Feature

How severe is the error you experienced?

☐ Minor - this issue has little or no impact on my ability to use BenMAP-CE

☒ Major - this issue significantly hinders my ability to use BenMAP-CE

☐ Blocking - this issue prevents me from using BenMAP-CE

What component of BenMAP-CE does the error affect?

Graphic User Interface: suggested changes

The audit trail report may help us to debug or understand your report better.

☒ Include BenMAP-CE generated audit trail report.

Please describe what you were doing when you encountered the error. Can you tell us how to reproduce the error? (5000 character limit)

To submit this report to us, click the Submit button. Thank you for your time.

Submit Cancel

3.3 Outputs

3.3.1 Results

If you are interested in viewing or exporting the results of an analysis, these reports can be accessed by clicking on the appropriate **Results** tabs from the upper portion of the main BenMAP-CE window. The number next to the tab description indicates the associated “step” of the BenMAP-CE analysis (from the tree menu).

Endpoint Group	Endpoint	Pollutant	Author	Start Age	End Age	Dataset Na...	Race
Acute Bronchitis	Acute Bronchitis	PM2.5	Docker...	8	12	EPA Standar...	

☒ Create map, data and chart (double-click the selected study)
 ☐ Create data (table) for multiple studies

Aggregation for raw data: ▼

Select Result Fields
Show Results

Select the appropriate tab for the type of results you wish to create:

- **Health Impact Results** uses a *Configuration Results* file (with the *.cfgrx* extension) to create a map, data table and bar chart for incidence results based on the selected health impact studies. “Raw” incidence estimates are those that have not been aggregated, pooled or valued. See Chapter 6: Estimating Incidence, for detailed instructions for creating health impact results.
- **Pooled Incidence Results** uses an *Aggregation, Pooling, and Valuation Results* file (with the *.apvrx* extension) to create maps, data tables and bar charts for incidence, aggregated incidence and pooled incidence results. See Chapter 7: Aggregation, Pooling, and Valuation, for detailed instructions for creating pooled incidence results.
- **Pooled Valuation Results** also uses an *Aggregation, Pooling, and Valuation Results* file (with the *.apvrx* extension) to create maps, data tables and bar charts for valuation, aggregated valuation and pooled valuation. See Chapter 7: Aggregate, Pool, and Value, for detailed instructions for creating pooled incidence results.

The results data are viewable on the **Data** tab in the lower right frame of the BenMAP-CE main window (see below). You can also view the results in a **GIS Map** (described in the following section), or simple **Chart** format. All results can be exported as comma-separated value files (.csv), which can be read into spreadsheet and database programs.